

Programme Code: TU856, TU857

Module Code: CMPU 3007

CRN: 22411, 22518

TECHNOLOGICAL UNIVERSITY DUBLIN

CITY CAMPUS - GRANGEGORMAN

TU856 – BSc. (Honours) in Computer Science
TU857 - Bachelor of Science (Honours) in Computer
Science (Infrastructure)

Year 3

SEMESTER 1 EXAMINATIONS 2024/25

CMPU 3007 Cloud Computing

Internal Examiner(s):

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The examination paper contains FOUR questions.
You must answer question 1 and any TWO questions from the
remaining three questions.

Please start each question on a new page.

Exam Duration: 2 hours

Question 1 – (Compulsory) [40 Marks]

Write a detailed explanatory note on all **FOUR** (4) of the following topics:
(10 marks each)

- (a) Network Attached Storage.
- (b) Cloud Security Concerns.
- (c) Benefits of Continuous Integration.
- (d) Considerations for avoiding Cloud Vendor lock-in.

Answer any TWO from Questions 2, 3 & 4

Question 2 [30 Marks]

- (a) Discuss and critically assess the following THREE (3) best practices of the Twelve Factor App from the point of view of a distributed Cloud native application:
 - i. Codebase. **(5 marks)**
 - ii. Dependencies. **(5 marks)**
 - iii. Configuration. **(5 marks)**
- (b) Describe in detail what is meant by each of these THREE (3) terms and illustrate your answers with an example:
 - i. Load Balancing. **(5 marks)**
 - ii. Elasticity. **(5 marks)**
 - iii. Auto Scaling. **(5 marks)**

Question 3 [30 Marks]

- (a) Compare and contrast the different architectural approaches between Docker Containers and Virtual Machines, highlighting the benefits and limitations of each approach. **(15 marks)**
- (b) There are four phases an organisation typically follows when considering a migration of its infrastructure to the Cloud. The final three phases are Proof of Concept; Migration; and Optimization. For each of these THREE (3) phases, discuss and critically appraise the details to be considered. **(3 x 5 = 15 marks)**

Question 4 [30 Marks]

DevOps transforms the end-to-end Software Development Life Cycle from a sequence of independent stand-alone stages to an integrated, seamless and automated pipeline.

- (a) Explain how such a transformation is achieved. **(6 marks)**
- (b) Discuss in detail the DevOps Best Practices. **(14 marks)**
- (c) Identify and explain the Benefits of a DevOps transformation. **(10 marks)**

<END OF EXAMINATION PAPER>